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ORIGINAL ARTICLE

Feasibility and accuracy of the screening for diabetic retinopathy using a fundus camera and an artificial intelligence pre-evaluation application

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Abstract

Aims Periodical screening for diabetic retinopathy (DR) by an ophthalmologist is expensive and demanding. Automated DR image evaluation with Artificial Intelligence tools may represent a clinical and cost-effective alternative for the detection of retinopathy. We aimed to evaluate the accuracy and reliability of a machine learning algorithm.

Methods This was an observational diagnostic precision study that compared human grader classification with that of DAIRET®, an algorithm nested in an electronic medical record powered by Retmarker SA. Retinal images were taken from 637 consecutive patients attending a routine annual diabetic visit between June 2021 and February 2023. They were manually graded by an ophthalmologist following the International Clinical Diabetic Retinopathy Severity Scale and the results were compared with those of the AI responses. The main outcome measures were screening performance, such as sensitivity and specificity and diagnostic accuracy by 95% confidence intervals.

Results The rate of cases classified as ungradable was 1.2%, a figure consistent with the literature. DAIRET® sensitivity in the detection of cases of referable DR (moderate and above, "sight-threatening" forms of retinopathy) was equal to 1 (100%). The specificity, that is the true negative rate of absence of DR, was 80 ± 0.04.

Conclusions DAIRET® achieved excellent sensitivity for referable retinopathy compared with that of human graders. This is undoubtedly the key finding of the study and translates into the certainty that no patient in need of the ophthalmologist is misdiagnosed as negative. It also had sufficient specificity to represent a cost-effective alternative to manual grade alone.

Keywords Diabetic Retinopathy Screening · Artificial Intelligence · Machine Learning · Sensitivity Specificity study · Ophthalmologist referral

Introduction

Since the 1920s, when insulin therapy transformed diabetes from a subacute fatal disease into a chronic encumbered disease, diabetic retinopathy (DR) has been the paradigm of long-term diabetic complications. In fact, DR is the most blood glucose-dependent complication and, despite new therapies, remains the most common cause of vision loss in the working-age population [1, 2]. Diabetologists have a crucial role to play in prevention, as the pursuit of optimal metabolic control is the most effective intervention to prevent retinal damage. Ophthalmologists also have a variety of treatments for diabetic retinopathy (laser, intravitreal injections, vitrectomy) that are highly effective in preventing vision loss [3, 4]. All diabetes guidelines at any level, national and international, whether type 1 or type 2,

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recommend regular early evaluation of retinal status and immediate intervention if necessary.

However, this is theoretical, as current estimates suggest that access to this type of screening in developed countries is low [5, 6]. Part of the reason why RD screening is so unsatisfactory lies in the complexity and cost of the procedures chosen for this task. In many countries [7], including Italy, referral to ophthalmologists, with the sole purpose of screening for RD, is still widespread, resulting in an excess of expensive and inappropriate referral requests. Lean telemedicine programmes for the screening of diabetic retinopathy, where diabetologists and ophthalmologists only play a supervisory role, are urgently needed, especially those in which the passage retinography/image/interpretation is alleviated by Artificial Intelligence (AI).

Thanks to 15 years of experience of retinal images taken by nurses with digital cameras and uploaded to a digital folder [8] for specialist evaluation, our team was well positioned to analyse a leaner procedure. DAIRET®, a screening algorithm, is the commercial Italian version of Retmarker Screening by Retmarker SA, Coimbra, Portugal. It was tested to filter out subjects that really needed to be referred to an ophthalmologist. Of course, when adopting an automatic screening method, the greatest concerns are about reliability and the risk of missing positive patients. The purpose of this study was to evaluate the diagnostic quality and accuracy of the DAIRET® app which is available (nested) in the Electronic Medical Record (EMR) MètaClinic by Meteda.

Methods

We conducted a diagnostic-accuracy study involving 4 diabetes and eye centres of 4 health districts in ASLTO5, a local health unit in the metropolitan area of Turin (Italy). Participants who were 18 years or older, with a diagnosis of any type of diabetes mellitus (excluded gestational), were enrolled in the DR screening at the time of their periodic visit for diabetes.

The main exclusion criteria were the presence of myopia > 6D, the presence of age-related macular degeneration, both in the early (drusen) and late forms, and the presence of any other macula disorder. These exclusion criteria were evaluated on the basis of previous ophthalmic examinations, mostly performed at the time of taking over by the centre.

All patients provided their informed written consent. Retinal images were obtained with a true colour, confocal, non-mydriatic fundus imaging system (DRSplus ® Centervue Spa, a company of iCare Finland Oy; Vantaa, Finland). Pictures were always snapped by nurses without pupil dilation. This fast and fully automated camera allows imaging through pupils as small as 1.8 mm, without the need for dilation, ensuring a comfortable patient experience and also offering the advantage of short examination time.

Two photographic image fields were taken of each eye, one centred on the optic disc and the other on the macula (Fig. 1). Subsequently, the images (the maximum capacity of the digital folder is 5 images for each visit) were uploaded to the Metaclinic EMR through a hospital network. DAIRET®, the machine learning algorithm based on artificial intelligence nested in MètaClinic, is capable of recognizing elementary lesions of diabetic retinopathy and issuing a first alert message. Retmarker Screening is a certified medical device, class II a in Europe, also approved by TGA Australia. Machine learning was trained in DR evaluation in 2015 and 2018 on more than 55,000 patients and 245,000 images. It should be noted that this algorithm does not provide a classification of lesions, but only a warning that a referral to a specialist is needed. The retinal images in each patient were graded by an ophthalmologist (human grader defined as Gold standard) and compared with artificial intelligence. The human grader followed the classification of the International Clinical Diabetic Retinopathy Severity Scale [9] (Table 1).

Results

Thirty-nine patients (5.7% of the sample) who had a history of severe myopia or macula disease were previously excluded according to the list of exclusion criteria. The percentage of cases classified as not evaluable by the human grader was 1.2%. These figures are in line with those reported in the literature, where the rates of low quality images range 4, 85–17, 3% [10, 11].

In the end, 637 patients were included, enrolled between June 2021 and February 2023. The main demographic characteristics are reported in Table 2. The mean age of the participants was 65 years (range 18–92 years), reflecting the advanced age of the population attending diabetes centres in Italy. 61 (9%) had type 1 diabetes, 555 (82%) type 2. 43% of the participants were women and 57% were men. As expected, subjects with DM1 were found to be younger but with a longer duration of the disease compared with DM2 patients.

Any type of retinopathy was present in 136 patients (21.3% of the total sample), mild DR in 101 patients, while moderate retinopathy or more (so-called sight-threatening DR) was found in 35 patients (5.5% of the overall sample). Table 3, labelled "No match", shows the characteristics of the 133 patients whose DAIRET® report does not match that of the Human Grader. Percentually this phenomenon was more frequent in DM1 subjects for which it was arranged that the ophthalmologist re-evaluated the cases 1 to 1. This verification allowed us to

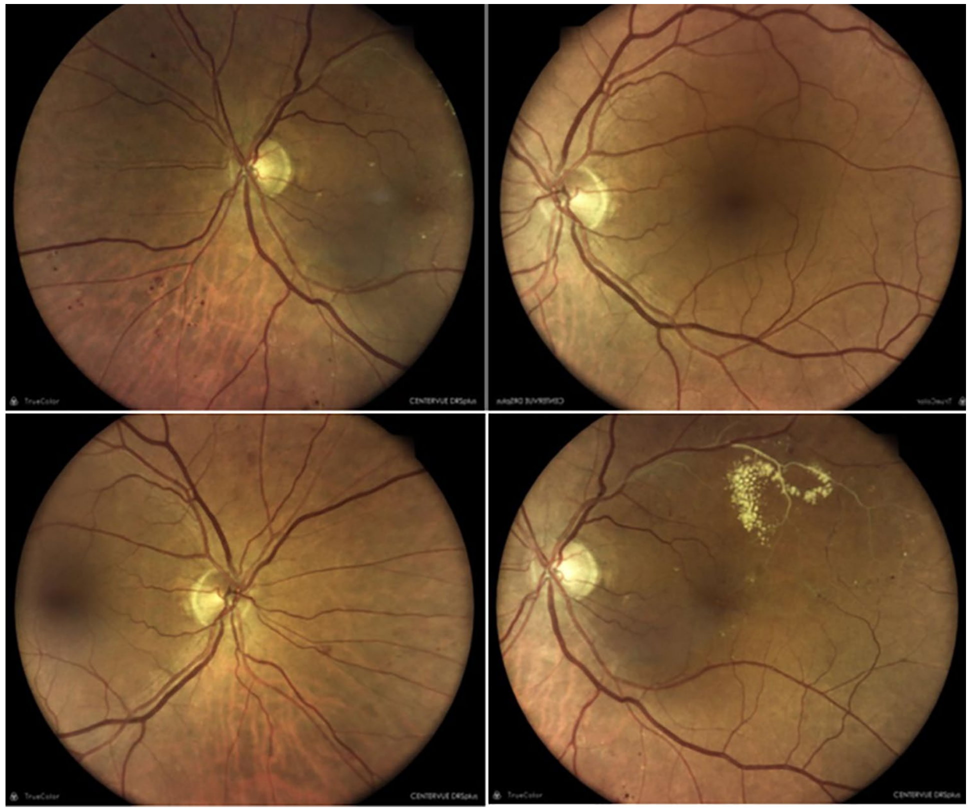

Fig. 1 Two photographic image field were taken of eye each one centred on the optic disc and the other on the macula

Table 1 International clinical diabetic retinopathy disease severity scale

Proposed disease severity level	Findings observable upon dilated ophthalmoscopy
No apparent retinopathy	No abnormalities
Mild NPDR	Microaneurysms only
Moderate NPDR	More than just microaneurysms but less severe NPDR
Severe NPDR	US Definition Any of the following (4-2-1 rule) and no signs of proliferative retinopathy: Severe intraretinal haemorrhages and microaneurysms in each of four quadrants Definite venous beading in 2 or more quadrants Prominent IRMA in 1 or more quadrants International Classification Any of the following and no signs of PDR More than 20 intraretinal haemorrhages in each of four quadrants Definite venous beading in two or more quadrants Prominent IRMA in one or more quadrants
PDR	One of both of the following: Neovascularization Vitreous/preretinal haemorrhage

NPDR nonproliferative diabetic retinopathy, *IRMA* intraretinal microvascular abnormalities, *PDR* proliferative diabetic retinopathy

Source: American Academy of Ophthalmology Retina-Vitreous Panel. Preferred Practice Pattern® Guidelines. *Diabetic Retinopathy*. San Francisco, CA: American Academy of Ophthalmology; 2014. Available at: www.aao.org/ppp identify 16 cases of foveal reflex (24% of type 1 patients), namely, physiologic reflex of light caused by the foveola retinae which, in young people, may lead AI to a wrong evaluation (false positive).

Key data from the results of this work are reported in Table 4. The sensitivity in detecting cases of moderate DR and above (referable) was equal to 1 (100%). Specificity, that is true negative rate for absence of DR, was 0.80 ± 0.04. The sensitivity to detect cases of mild DR, R1, was equal to 0.69 ± 0.09. We also performed a Confusion Matrix which confirmed these figures (Table 1 additional online material). Finally, after exclusion of type1 patients, a sensitivity analysis of patients above or below the median age (67 years) confirmed that the sensitivity remains the same and only the specificity for the absence of retinopathy is slightly worse in older patients. See Table 2 additional online material.

Discussion

By identifying only one reading centre with an ophthalmologist chosen as the gold standard and implementing the experimental algorithm DAIRET® within the EMR used for daily clinical activity, we were able to perform an accuracy study on a large sample of patient retinograms.

This precision study is based on the need to find streamlined and cost-contained procedures to implement retinopathy screening in large populations such as individuals with diabetes. To reduce the screening burden on ophthalmologists, it is desirable to develop automated image assessment systems, leaving the task of treatment and taking care of pathological cases to specialists. The use of apps such as DAIRET® to quickly arrive at a definition of patients as referable or not referable can significantly affect the efficiency and effectiveness of this important screening. With the results of this work on 637 subjects, we are able to send

Table 2 Characteristics of the study population

Patients	Total (%)	Age (years) ± SD	Range (years)	Duration of DM (years) ± SD
All patients	637 (100)	65 ± 12.7	18–92	13 ± 9.5
Men	383 (60)	64 ± 12.7	18–89	13 ± 9.5
Women	254 (40)	66 ± 12.8	32–94	13 ± 9.5
DM1	61 (10)	48 ± 12.8	18–80	22 ± 9.5
DM2	555 (87)	67 ± 12.7	34–92	12 ± 9.5
Other types of DM	21 (3)	50 ± 12,7	32–71	8 ± 9.5

Table 3 Patients whose AI diagnosis does not match that of human grader (FN + FP)

No match*	Total (%)	Age (years) ± SD	Range (years)	Duration of DM (years) ± SD
No match* in all patients	133 (21)	60 ± 12.8	18–89	14 ± 9.5
No match in Men	83 (22)	59 ± 12.8	18–89	15 ± 9.5
No match in Women	50 (20)	61 ± 12.7	32–89	14 ± 9.5
No match in DM1	30 (45)	42 ± 12.8	18–80	22 ± 9.5
No match in DM2	99 (18)	66 ± 12.7	34–89	13 ± 9.5
No match in other types of DM	4 (18)	42 ± 12.7	32–53	9 ± 9.4

*Patients whose AI diagnosis does not match that of human grader (FN + FP)

Table 4 Key results of the work with sensitivity and specificity of the precision analysis

Retinopathy grade (by manual grader)	Human grader	Dairet	Dairet Specificity (IC 95%)	FP (%)
No apparent DR	501	399	0,80 (IC 95% ± 0,04)	102 (20,4)
Retinopathy grade (by manual grader)	Human grader	Dairet	Dairet Sensitivity (IC 95%)	FN (%)
Mild DR	101	70	0,69 (IC 95% ± 0,09)	31 (30,7)
Moderate DR and beyond	35	35	1 (100%)	0 (100)

a strong message on the reliability and safety issue of the DAIRET® system, in that the specificity for moderate and above forms, of DR, is 100%. That is, there is the certainty that no patient in need of the ophthalmologist is misdiagnosed as negative and sent home. This is undoubtedly the key finding of the study, as it sheds a light of safety on the system.

Sensitivity for mild forms of DR is limited to 69% and this might appear to be a disappointing figure. In reality, in these early forms, characterized only by microaneurysms, the ophthalmologist is not directly concerned since the therapy of choice is to correct metabolic control as best as possible, a field in which the diabetologist is mainly involved and responsible. The proof of this concept is that throughout the literature on retinal screening in diabetes, the safety threshold is identified as that involving ophthalmologist intervention, namely moderate forms and up.

DAIRET® achieved an acceptable specificity of 80%, that is, 1 in 5 patients erroneously test positive for the presence of any form of retinopathy. This figure can be improved over time by self-learning ML systems and reducing unnecessary referrals. This limitation is in part due to the foveal reflex, which is responsible for false positive cases in young patients with type 1 diabetes and which may be amplified by some type of non-mydriatic camera. This problem can be improved in the algorithm by technicians and, in part, in self-learning by the software itself with a better integration between the device and the automated assessment system.

These results are encouraging compared to the data from the literature on retinal reading systems with Ai. In general, our sensitivity for referable DR of DAIRET® is better than other tests, while specificity is similar. The study that led to a landmark FDA approval, as the first fully autonomous AI diagnostic system [12], was based on the ability to detect more than mild DR in two field 45° fundus images and, overall, the sensitivity was found to be 87.2%, with a specificity of 90.7%.

We report for comparison the latest sensitivity and specificity data of IDX-DR (IDx LLC, Iowa, USA (91% and 84%) [13], of EYEART app (91.7% and 91.5%) [13, 14], and of RETMARKER app (sensitivity 85%) [15].

As a safety measure, retinal examination should not be used in the first evaluation of ocular complications, as baseline information on other eye pathologies is of great importance. However, considering the pressing need for mass screening in some countries, the decision should be made on the basis of the local organization and needs.

The results of this work pave the way for daily practical use of DAIRET® with marked simplification and time savings for both the patient and the service. In particular, it should be emphasized that DAIRET® is managed within the MètaClinic electronic medical record: this characteristic allows the physician to have all of the patient's clinical data available at the click of a button and, if necessary, to prescribe corrections to diabetologic and ophthalmologic therapy. As regards data protection regulations, the fact that data are locally elaborated and not sent to an external iCloud is a relevant strength.

Widespread telemedicine programmes could be developed with the potential to distribute quality eye care to virtually any location and address the lack of access to eye care, while currently too many subjects are excluded from the advances achieved during the past decades [16]. However, one of the main barriers to a wider implementation of teleophthalmology has remained the requirement and costs of trained personnel for image reading and classification of a large volume of retinal images [17, 18]: these are the key points where an app such as DAIRET® can make a difference.

The main limitation of this study is the use of only one ophthalmologist as human grader. Surely dual reporting would have improved the definition of the gold standard. On the other hand a strength of this real-world study lies in the execution of screening performed by professional figures from diabetes teams, especially nurses, which lends an image of feasibility and reality to the trial. Nurses minimally trained can easily acquire retinal images and upload them to a digital folder, in this way, monitoring of retina complications may be performed by non-eye care professionals. Nevertheless, future research is required to address several challenges of automated image detection algorithms [19]: for example, medicolegal implications or management of other ophthalmic disorders like age macular degeneration, often detected in diabetic patients undergoing screening for retinopathy.

Conclusions

AI-driven screenings can significantly lower healthcare costs by reducing the need for specialized human resources and making the screening process more efficient. While AI has shown great promise in diabetic retinopathy screening, it's essential to note that these systems should be seen as tools to aid, not replace, human healthcare professionals. AI models are only as good as the data they're trained on, and there may be rare or complex cases where human expertise is necessary. Additionally, patients diagnosed with diabetic retinopathy will need human healthcare professionals for treatment and ongoing management of their condition. AI in healthcare, especially in diabetic retinopathy screening, represents an exciting frontier in medicine. By combining the strengths of AI and human expertise, we can hope for more accurate, accessible, and effective healthcare in the future. We conclude that screening examinations or any tests performed in medicine should only be performed when the results have a reasonable probability of altering the treatment and when the risk-to-benefit ratio of undergoing the test is favourable for the individual patient. In this work, DAIRET® screening has been shown to meet both requirements.

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Declarations

Conflict of interest The authors declare that they have no conflicts of interest.

Ethical standard statement The study was approved by the Local Ethics Committee.

Informed consent All patients signed an informed consent disclosure.

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